

Model Evaluation

```
[108]: from sklearn.metrics import mean_absolute_error

mae = mean_absolute_error(y_test, y_pred)

print("MAE:", mae)

MAE: 50898.7395349408
```

```
[110]: # Next:MSE – Mean Squared Error
```

```
[111]: from sklearn.metrics import mean_squared_error

mse = mean_squared_error(y_test, y_pred)

print("MSE:", mse)

MSE: 5285482803.266442
```

RMSE

```
[112]: from sklearn.metrics import root_mean_squared_error

rmse = root_mean_squared_error(y_test, y_pred)

print("RMSE:", rmse)

RMSE: 72701.32600762136
```

Now the final major metric: R^2

```
[114]: from sklearn.metrics import r2_score

r2 = r2_score(y_test, y_pred)

print("R²:", r2)

R²: 0.5966539537632211
```

✅ Approximately 59.7% of the variance in median_house_value is explained by the model on the test set.



Next — Evaluate Model #2

```
[88]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

mae_2 = mean_absolute_error(y_test, y_pred_2)
mse_2 = mean_squared_error(y_test, y_pred_2)
rmse_2 = np.sqrt(mse_2)
r2_2 = r2_score(y_test, y_pred_2)

print("Model #2 - Decision Tree Regression")
print("MAE :", mae_2)
print("MSE :", mse_2)
print("RMSE :", rmse_2)
print("R² :", r2_2)

Model #2 - Decision Tree Regression
MAE : 43803.50581395349
MSE : 5120584623.200582
RMSE : 71558.26034218958
R² : 0.6092376724956141
```



Next — Evaluate Model #3

```
[93]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

mae_3 = mean_absolute_error(y_test, y_pred_3)
mse_3 = mean_squared_error(y_test, y_pred_3)
rmse_3 = np.sqrt(mse_3)
r2_3 = r2_score(y_test, y_pred_3)

print("Model #3 - Random Forest Regression")
print("MAE :", mae_3)
print("MSE :", mse_3)
print("RMSE:", rmse_3)
print("R²  :", r2_3)
```

```
Model #3 - Random Forest Regression
MAE : 32325.262456395347
MSE : 2532578201.922123
RMSE: 50324.72753947231
R²  : 0.8067337568671218
```



Winner: Random Forest Regression

It wins on all three primary metrics:

Lowest MAE → predictions are closest on average. Lowest RMSE → substantially fewer/severer large errors. Highest R^2 → explains the largest proportion of variation.

So our decision is:

Random Forest Regression is the best candidate among the three models we tested. ✅